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- **Problem Statement ID –** SIH26149
- **Problem Statement Title –** Design and Development of an Integrated Secure Data Erasure and Advanced File Recovery Tool for Digital Forensics and Data Sanitization
- **Theme –** Blockchain & Cybersecurity
- **PS Category –** Software
- **Team ID –**
- **Team Name –** Paper Rex

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SANITIZEX

A unified 3-module platform for secure data sanitization and forensic file recovery.

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Secure Drive Eraser

Securely wipes entire storage devices such as HDDs, SSDs, USB drives and memory cards, with erasure verification, audit logs and tamper-resistant reports.

01

Secure File & Folder Eraser

Enables selective deletion of files/folders while cleansing associated metadata and residual traces, with batch operations, verification and audit reporting.

02

Advanced File Carving & Recovery

Recovers formatted, damaged or corrupted media using signature and structure-based carving techniques, including fragment reconstruction and confidence scoring.

03

Problem	Multiple separate tools for erasure & recovery	Limited support across storage technologies & filesystems	Difficult to prove that data was actually erased
Our Fix	**One integrated platform for both operations**	**Multi-device & multi-filesystem support**	**Verification + audit logs + forensic reports for traceability and accountability**

Uniqueness

1 SURGICAL ERASURE

File-level precision down to its physical blocks + metadata

2 HARDWARE-AWARE

Right erasure mechanism for HDD, SSD & NVMe

3 SAFETY BY DESIGN

Recovery is read-only at both API and OS levels

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TECHNICAL APPROACH

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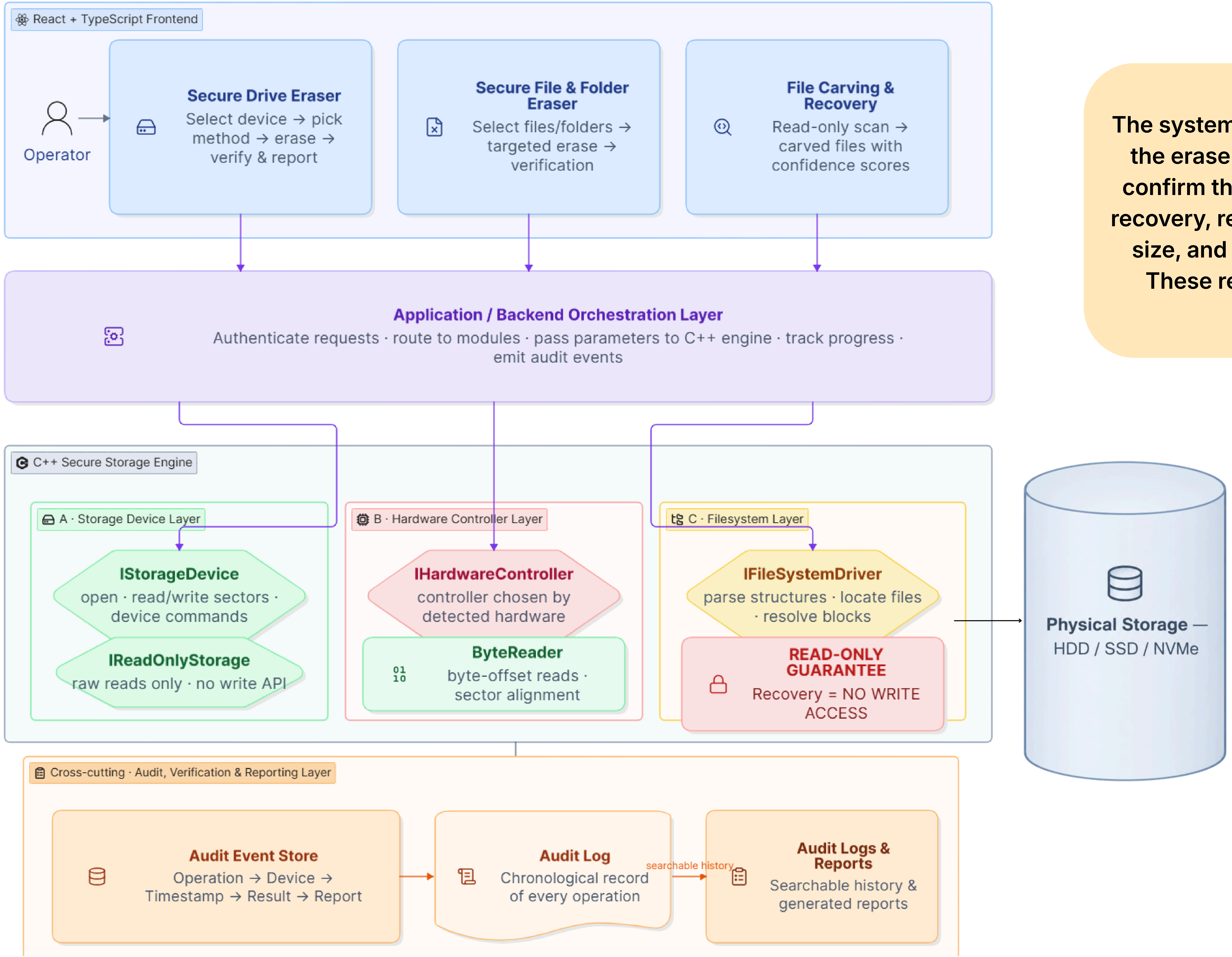

The system verifies erasure by auditing data before and after the erase or wipe operation and comparing checksums to confirm that the original content has been removed. During recovery, recovered files are checked for structure, type, and size, and a SHA-256 checksum is generated for each file. These results are recorded to confirm the integrity and reliability of the recovery process.

Tech Stack

WINDOWS API

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FEASIBILITY AND VIABILITY

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Feasibility Areas

Technical
Built on low-level OS I/O with modular interfaces for storage, hardware and filesystem operations

Working Prototype
Read-only storage access and MBR/GPT partition detection are already implemented and tested.

Filesystem Support
ext4 erasure and recovery support has been implemented, including inode, extent and metadata handling.

Safety
Recovery is protected by read-only APIs + OS-level read-only handles, preventing write operations by design.

Extensibility
The layered architecture allows additional devices, operating systems and filesystems to be integrated independently

Potential Challenges/ Risks	Hardware-specific limitations	Large-scale data processing	Filesystem edge cases	Compliance & evidential integrity
Strategy to Overcome	**Detect device capabilities and select the safest supported erasure method.**	**Use chunked reads/writes and optimized low-level I/O to manage memory and performance.**	**Validate filesystem structures and use fallback handling for unsupported/corrupt cases.**	**Maintain verifiable logs, reports and operation records for every action.**

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IMPACT AND BENEFITS

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Impact On

Government & Defence Agencies

Safer disposal of sensitive data and improved compliance with sanitization standards.

Enterprises & Organizations

Reduces data-leak risks while simplifying secure device disposal and auditing.

Law Enforcement & Forensic Teams

Faster investigation through advanced recovery and structured forensic reporting.

Cybersecurity Professionals

One platform for sanitization + recovery, reducing tool dependency and investigation overhead.

Benefits

Social

Reduces the risk of sensitive data leaks and unauthorized recovery.

Economic

Reduces dependency on multiple specialized tools, lowering operational and investigation costs.

Legal

Verification, audit logs and forensic reports improve accountability and compliance.

Environmental

Enables secure reuse or disposal of storage devices without unnecessary destruction of usable hardware.

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RESEARCH AND REFERENCES

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